movement requires an expenditure of energy released by respiration (ATP). Thus, there is a link between water absorption and respiration. It is evident from the fact that when respiratory inhibitors like KCN, Chloroform are applied there is a decrease in the rate of respiration and also the rate of absorption of water.

2. Passive Absorption

In passive absorption, roots do not play any role in the absorption of water and is regulated by transpiration only. Due to transpiration, water is lost from leaf cells along with a drop in turgor pressure. It increases DPD in leaf cells and leads to withdrawal of water from adjacent xylem cells. In xylem, a tension is developed and is transmitted downward up to root resulting in the absorption of water from the soil.

In passive absorption (Table 11.3), the path of water may be symplastic or apoplastic. It accounts for about 98% of the total water uptake by plants.

Table: 11.3 Differences between Active Absorption and Passive Absorption

Active absorption	Passive absorption
Active absorption takes place by the activity of root and root hairs	The pressure for absorption is not developed in roots and hence roots play passive role
Transpiration has no effect on active absorption	Absorption regulated by transpiration
The root hairs have high DPD as compared to soil solution and therefore water is taken by tension	The absorption occurs due to tension created in xylem sap by transpiration pull, thus water is sucked in by the tension
Respiratory energy needed	Respiratory energy not required
It involves symplastic movement of water	Both symplast and apoplast movement of water involved

11.5 Ascent of Sap

In the last chapter, we studied about water absorption from roots to xylem in a lateral direction and here we will learn about the mechanism of distribution of water inside the plant. Like tributaries join together to form a river, millions of root hairs conduct a small amount of water and confluence in xylem, the superhighway of water conduction. Xylem handles a large amount of water to conduct to many parts in an upward direction.

The water within the xylem along with dissolved minerals from roots is called **sap** and its upward transport is called **ascent of sap.**

11.5.1 The Path of Ascent of Sap

There is no doubt; water travels up along the vascular tissue. But vascular tissue has two components namely Xylem and Phloem. Of these two, which is responsible for the ascent of sap? The following experiment will prove that xylem is the only element through which water moves up.

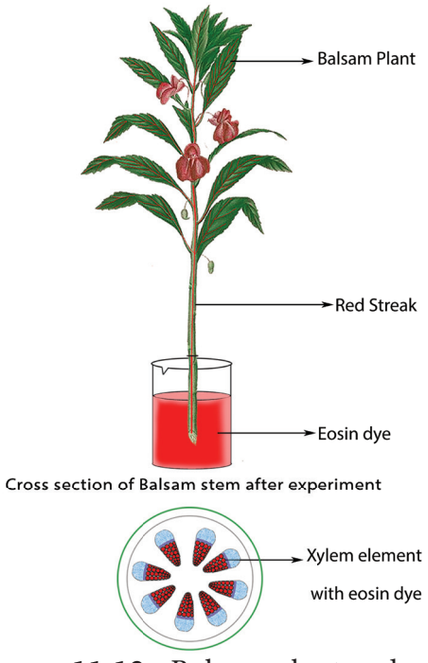


Figure 11.12: Balsam plant and eosin dye experiment

Cut a branch of balsam plant and place it in a beaker containing **eosin** (red colour dye) water. After some time, a red streak appears on the stem indicating the ascent of water. Remove the plant from water and cut a transverse

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